The Possibility Engine

Executive brief · AGI ALPHA FLYWHEEL

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28 October 2026

The possibility

A laboratory should be able to ask what instrument it needs to construct to answer a question that its existing measurements cannot resolve. An enterprise built around that ability would expand what its beneficiaries can investigate, rather than only make an existing procedure faster.

This campaign constructs a precise computational example. Two different hidden diffusion networks produce the same response at the existing port for every input waveform. Adding actuators alone does not help. Adding sensors alone does not help. A joint actuator–sensor addition makes the distinction robustly answerable.

Decision: retain the instrument; stop the tested narrow paid offering and redesign its application bridge. The capability change is demonstrated within the declared mathematical family. A commercially consequential physical flagship is not established.

What was actually built

The runnable constructor accepts a finite family of validated sixteen-node networks, allowed access locations, experiment costs, parameter bounds and a failure requirement. It produces a minimum-cost experiment or portfolio, explicit uncertainty ranges, an independently checkable witness and executable RC circuit inputs.

At the declared primary bounds, the output intervals are separated by **0.03633036 normalized units**, including measurement error. The original and both single-sided expansions remain uninformative. A successor task produces a **five-experiment portfolio costing 66 units** that distinguishes six hypotheses after access to any one terminal is lost; the internal network remains intact. These are finite design costs, not hardware quotations.

Fresh evidence	Result
Technical cases and malformed-input controls	42/42 pass
Binary constructions / required refusals	12 / 12
Fresh six-hypothesis successor portfolios	12/12 independently optimal
Exported circuits checked by ODE and ngspice	336/336 pass
Maximum ngspice discrepancy / tolerance	8.59×10^{-9} / 10^{-6}
Ordinary-library successor comparisons	12/12 exact ties

The graphs and theoretical principles have established prior art. An equally informed conventional constructor obtains the same useful result. Ordinary-library parity establishes portability, not exclusive retained advantage. No trained specialist or physical device is claimed.

Commercial delivery and complete economics

Simulation scope: all buyers, contracts, commercial acceptance, collections and enterprise results are simulated. Technical construction and circuit-software checks are executed. Costs and demand are disclosed estimates; they are not invoices or observed market behavior.

The candidate offering is a fixed-scope Computational Experiment Construction Pack for a prototype team that can still modify access. Its current implementation supports the tested regular-graph family, not arbitrary thermal systems. A buyer who only needs an equivalent thermal model can choose existing tools or self-delivery.

Two bounded price grids tested 24 policy entries, representing 20 unique combinations. The selected attempt quotes **USD4,500**, takes a **50% restricted deposit**, and allows **two new contracts per month**. Fresh confirmation executes 128 histories in each of ten scenarios, plus conventional and stop counterparts. Independent reconstruction reconciles **1,536 trajectories, 385,682 journal events and 1,828 contracts**, with no terminal deposits, liabilities or receivables left open.

Reference modeled result	USD
Complete construction cost before finance/exit	36,200
Mean recognized revenue	6,648.82
Mean operating contribution	−15,887.20
Mean full-project profit	−53,187.20
5th–95th world percentile profit	−57,403.12 to −49,832.99
Mean peak funding requirement	53,877.96
Stop after completed research and archive	−36,950.00
Additional expected loss from continuing	16,237.20

All 128 histories lose money in each modeled scenario. With the strongest free substitute available, the paid offering obtains zero sales. These are conditional simulation estimates, not probabilities measured in a real market. The reference mean's sampling standard error is USD231.12; uncertainty about the underlying demand assumptions is much larger and was not inferred from sampling precision.

What the successor should attempt

The next useful construction would connect the exact instrument to a beneficiary decision and a permitted physical fixture. Require a question whose answer changes an action; independently specified nonuniform-component, timing and calibration limits; an incomplete-hypothesis challenge; and a competent unrestricted specialist comparison. Then establish willingness to pay and complete delivery effort.

Expansion is staged. The modeled USD4,000 successor experiment does not pass the conservative incremental-value gate and receives no funding. The possible later USD20,000 integration stage remains separate. Shared commissioning can reduce duplicated work, but the same conventional arrangement receives that benefit; the simplified twelve-sponsor funding threshold is not evidence that twelve willing sponsors exist.

USD1 billion of ten-year incremental beneficiary value would require, at an assumed USD2,500 net value per accepted pack, **400,000 useful packs** and roughly **10,000 active beneficiaries** buying four annually. At current assumed capacity that implies about **1,667 delivery teams**. Reachable demand and that net value have not been established. Revenue, profit and valuation remain distinct.

Handoff

The complete paper, LaTeX, self-contained reading edition, runnable constructor, frozen protocols, raw confirmation, independent checks, complete simulated journals and preserved failures are included in the companion. The exact scientific freeze is `e1ccee55bf1269fd3fc0932538782a2eaaa1eeda31cf06880435708fd74e33d7`.

The contribution is a reproducible instance of complementary means changing answerability. The next burden of proof is a real productive application that survives equally equipped alternatives.

Find your Alpha. Build your successor Ω. Invent what becomes possible.